04-09-2025

$\tau w N = b(1-N)$

$\Pi_{t+1}' = \Pi_t' P$

$\hookrightarrow$ State $\times$ transition matrix

$\Pi_{t+1} = \Pi_t = \Pi$

$$(\Pi_e \;\; \Pi_u) = (\Pi_e \;\; \Pi_u) \cdot \begin{pmatrix} P_{ee} & P_{eu} \\ P_{ue} & P_{uu} \end{pmatrix}$$

$$\Pi_e = \Pi_e \cdot P_{ee} + \Pi_u + P_{ue}$$

$$\Pi_e (1 - P_{ee}) = (1 - \Pi_e) P_{ue}$$

$$\Pi_e (1 - P_{ee} + P_{ue}) = P_{ue}$$

$$\underset{N}{\Pi_e} = \frac{P_{ue}}{1 - P_{ee} + P_{ue}} = \frac{0.5}{1 - 0.96 + 0.5} = 0.92 = N$$

$$0.08 = (1-N) = \Pi_u$$

$$\tau = \frac{b(1-N)}{wN}$$

Steps for Solving:

1: compute invariant distribution $(\Pi_e \;\; \Pi_u)$ + find $N = \Pi_e = P_{ue}/1 - P_{ee} + P_{ue}$

2: guess r

3: compute $\left(\frac{K}{N}\right)$ from FOC for K, find w w/ $\left(\frac{K}{N}\right)$

4: use gov't budget balance to find $\tau = b(1-N)/wN$

5: given $\{w, r\}$ and $\{\tau, b\}$, solve consumer optimization problem and find c, a', v

Steps for Solving (cont.):

6: compute invariant dist. of agents over the state space; $f(a,e)$ and $f(a,u)$

7: compute new agg. capital:

$K_{new} = \int g(a,e) f(a,e)\,da + \int g(a,u) f(a,u)\,da$

Need policy $f(\cdot)$ and dist of people over states

8: Compute new interest rate

$$r_{new} = \alpha \left(\frac{K_{new}}{N} \right)^{\alpha - 1} - \delta$$

9: compare r and r_{new}:

if $r \approx r_{new} \Rightarrow$ done

else $r = \text{weight} \cdot r + (1 - \text{weight})\, r_{new} \Rightarrow$ back to 3

Invariant Dist: Non-Stochastic Sim:

$f(a, \varepsilon)_{m \times 2}$ (all start with K^{rep})

disc: $a_1 \dots a_m$, set $a^s = K^{rep}$

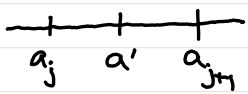
1: Start by guessing $f^{(0)}(a, \varepsilon)$

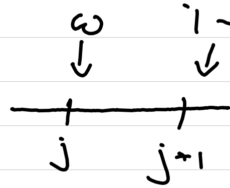
$$f^{(0)}(a,\varepsilon) = \begin{array}{c|cccccc} e & 0 & 0 & \dots & N & \dots & 0 \\ u & 0 & 0 & \dots & 1-N & \dots & 0 \end{array}$$

2: $f^{(1)}(a, \varepsilon) = 0 \ \forall\, a, \varepsilon$

\begin{tikzcd}
 & f^{(0)}(a_i, e) \arrow[dl] \arrow[dr] & \\
P_{ee} \arrow[d] & & P_{eu} \arrow[d] \\
e & & u
\end{tikzcd}

$$a' = g(a_i, e)$$





$$\omega = \frac{a_{j+1} - a'}{a_{j+1} - a_j}$$

weight savings so some is allocated to either potential state

$$f^{(0)} \quad \begin{array}{c|c} & a_1 \; \dots \; a_i \; \dots \; a_m \\ \hline e & \begin{pmatrix} \quad\quad \bullet \quad\quad \\ \\ \end{pmatrix} \\ u & \end{array}$$

$$f^{(1)} \quad \begin{array}{c|cc} & a_1 \dots a_j & a_{j+1} \dots a_m \\ \hline e & \dots P_{ee}\omega & P_{ee}(1-\omega) \dots \\ u & \dots P_{eu}\omega & P_{eu}(1-\omega) \dots \end{array}$$

\begin{tikzcd}
 & & f^{(0)}(a_i, e) \arrow[dll] \arrow[dl] \arrow[dr] \arrow[drr] & & \\
P_{ee} \cdot \omega \; f^{(1)}(a_j, e) & P_{ee}(1-\omega) \; f^{(1)}(a_{j+1}, e) & & P_{eu}\,\omega \; f^{(1)}(a_j, u) & P_{eu}(1-\omega) \; f^{(1)}(a_{j+1}, u)
\end{tikzcd}

Coding:

1: Initialize $f^{(1)} = 0$

2: Loop over current states $\varepsilon = [e, u]$, $a = [a_1, \dots, a_m]$

↳ for each combo, find savings $a' = g(a_i, \varepsilon)$ AND locate on grid

$$\omega = \frac{a_{j+1} - a'}{a_{j+1} - a_j}$$

↳ loop over future states ε'

$$f^{(1)}(j, \varepsilon) = f^{(1)}(j, \varepsilon') + f^{(0)}(i, \varepsilon) P(\varepsilon' | \varepsilon) \cdot \omega$$

$$f^{(1)}(j+1, \varepsilon) = f^{(1)}(j+1, \varepsilon') + f^{(0)}(i, \varepsilon) P(\varepsilon' | \varepsilon) \cdot (1 - \omega)$$

↳ end loop for ε'

end loop for (a_i, ε)

Comare $f^{(0)}$ and $f^{(1)}$

$f^{(0)} = f^{(1)}$: done

$f^{(0)} \neq f^{(1)} \Rightarrow$ set $f^{(0)} = f^{(1)}$; $f^{(1)} = 0$

Monte-Carlo Simulations:

1: Choose (large) sample size

2: Init sample

↳ assign each agent initial state variables

i.e. give all K^{rep} and draw ε from invariant dist

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Draw RND
   if RND < N
      ε = e;
   else
      ε = u;
   end
```

3: Compute set of moments to determine convergence

↳ $\text{moments}^{(0)} = [\bar{N}, \bar{a}, b_a]$

4: Compute next per. assets + employment:
$a^{(1),i} = g(a^{(0),i}, \varepsilon^{(0),i})$, draw uniform random number RND
compare with $P(\varepsilon^{(1)} | \varepsilon^{(0),i})$

5: Recompute set of moments$^{(1)}$

6: Compare $M^{(0)}$ and $M^{(1)}$ until converge